

Hafiz Tanzeel Shamshad

AI/ML Engineer | Web Developer

AI/ML Engineer & MERN Stack Developer with experience in web development, machine learning, Deep learning and computer vision. Skilled in React, Node.js, Express, and MongoDB, with 1 year of hands-on experience in building a dental disease detection system for my FYP project, applying computer vision and advanced ML techniques.

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👤 Hafiz-Tanzeel-Shamshad

WORK EXPERIENCE

Lab Assistant

Riphah International University
Jan 2024 - Dec 2025 | Islamabad

Responsibilities & Contributions

- Worked as a Lab Assistant, helping students understand key concepts in Programming Fundamentals, Assembly Language, and Artificial Intelligence.
- Assisted in practical labs and guided students through assignments.
- Supported faculty by preparing lab materials and ensuring smooth class execution.

Python Developer Intern

TMR Consulting
Microsoft Gold Partner and Tier 1 Cloud Solution Provider.
Jan 2024 - Mar 2024 | Islamabad

Responsibilities & Contributions

- Developed and maintained backend features using Python and Django.
- Built RESTful APIs using Django REST Framework for data retrieval, user management, and authentication modules.
- Implemented CRUD operations and improved database performance.

Frontend Developer Intern

Kairiz Cyber Technologies
A tech company specializing in digital solutions, and web development.
Jun 2024 - Aug 2024 | Islamabad

Responsibilities & Contributions

- Developed interactive user interfaces using React.js, JavaScript, and modern frontend development practices.
- Built reusable functional components with Hooks (useState,useEffect, useContext) to manage UI state and application logic.
- Integrated REST APIs into React applications using fetch or Axios, displaying dynamic and real-time data.

EDUCATION

Bachelor of Science in Computer Science

Riphah International University
Feb 2022 - Feb 2026 | Islamabad | CGPA: 3.86/4.00

SKILLS

Javascript

React.js

Redux Toolkit

Tailwind CSS

Bootstrap

Node.js

Express.js

RESTful APIs

MongoDB

MySQL

Python

Numpy

Pandas

Scikit-learn

OpenCV

Computer Vision

GitHub

PROJECTS

Dental Disease Detection System

Nov 2024 - Dec 2025

Tech Stack: React.js (Web App), Flutter (Mobile App), FastAPI, Supabase, TensorFlow, YOLOv8l & YOLOv9c, OpenCV, Clickup

Key Features

- Implemented a system for detecting dental caries and periodontal disease using TensorFlow/Keras, achieving 88% accuracy.
- Applied OpenCV-based preprocessing (noise reduction,contrast enhancement, segmentation) to boost model performance.
- Integrated real-time model inference, enabling users to upload dental images and instantly view detection results with bounding boxes and confidence scores.
- Created a user interface for visualizing results and documented the comprehensive research and model workflows.

BuyNow Web Application [🔗](#)

Nov 2025 - Dec 2025

Tech Stack: React.js, Redux, Tailwind, Node.js, Express.js, MongoDB, JWT, Axios.

Key Features

- JWT-based authentication with access/refresh tokens and role-based access control.
- Build an Admin dashboard for managing users, products, and orders, as well as product browsing and search based on name.
- Payment options: Cash on Delivery and mock card payments.

Airbnb Booking Application [🔗](#)

Jun 2022 - Sep 2022

Tech Stack: Node.js, MongoDB, EJS, Passport.js, Cloudinary, Mapbox, Render.

Key Features

- Built with EJS templating for SEO optimized server side rendering.
- Integrated Cloudinary for image management & Mapbox API for interactive property maps.
- Developed CRUD operations for property listings.